Theoretical Spectroscopy via the Cumulant Expansion

Patryk Kozlowski

April 13, 2026

Outline

1. Introduction
2. Methods and results
3. Future directions

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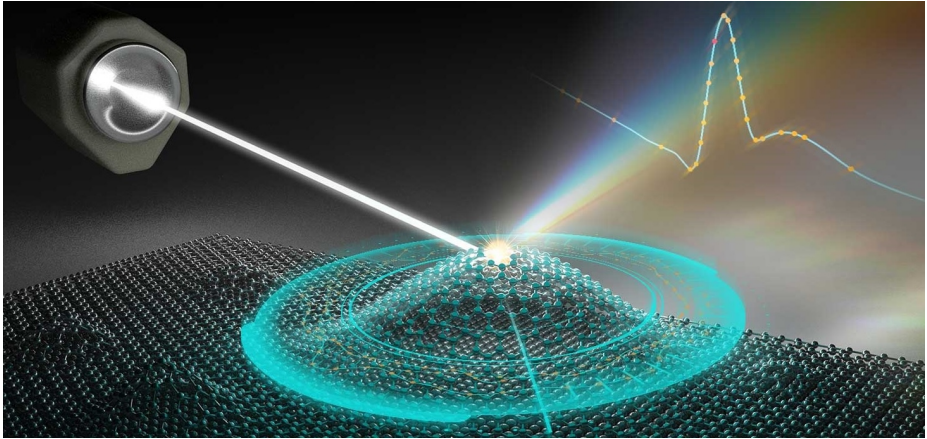

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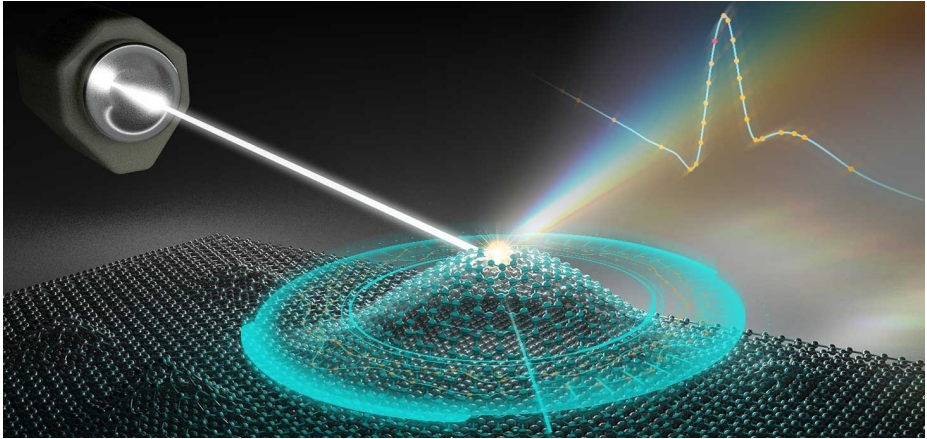

- The spectral function $A_{pp}(\omega)$ for a core state p is the observable in experiments

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- I consider several approaches for obtaining Im $G^R_{pp}(\omega) \propto A_{pp}(\omega)$

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- I explore three different approximations for $\Sigma^R_{pp}(\omega)$: GW, RSPT2, and a size-consistent BWPT2.

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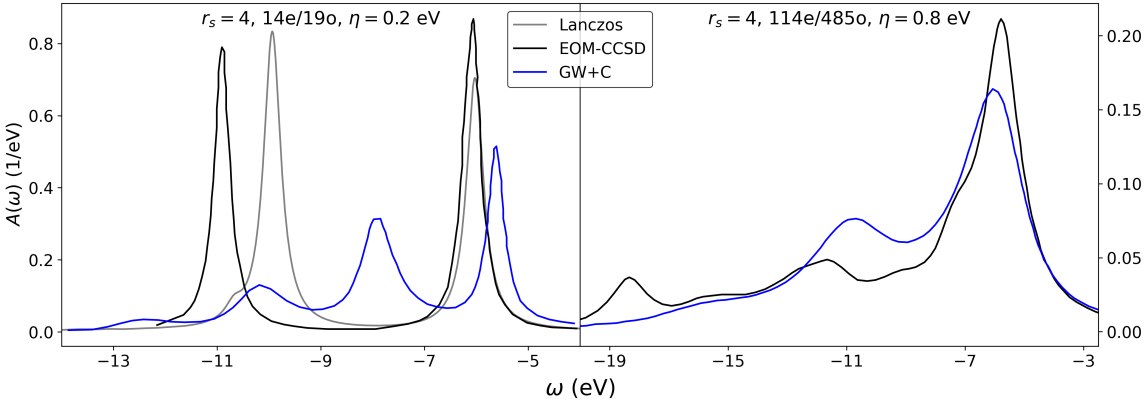

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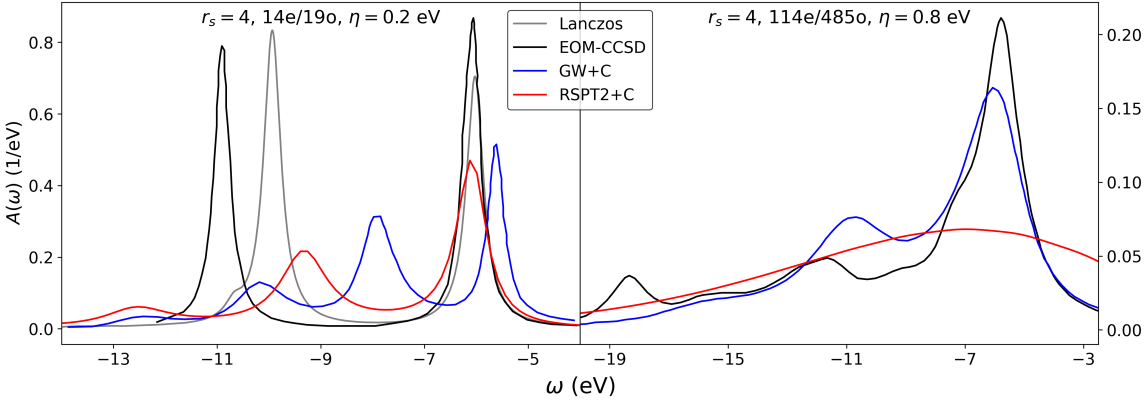

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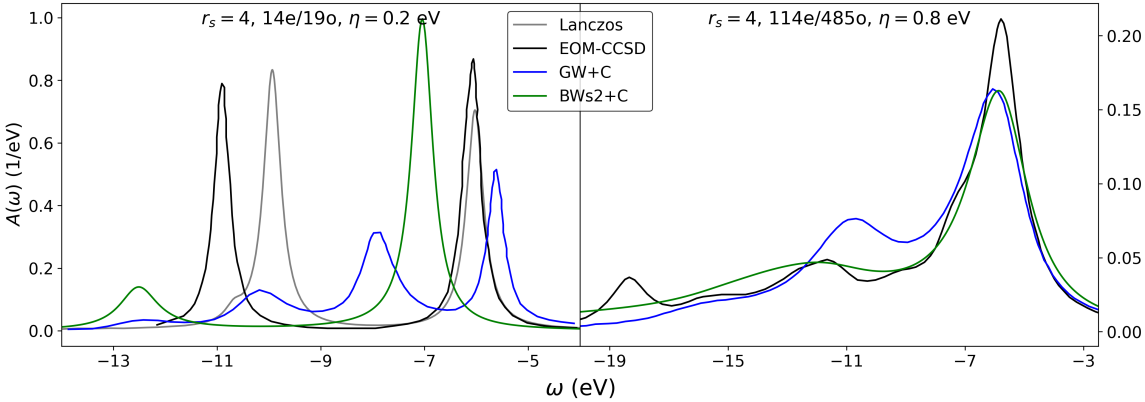

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using the cumulant as

$$C^R_{pp}(t) = -\frac{1}{\pi} \int d\omega \frac{\mathrm{Im}\Sigma^R_{pp}(\omega)}{\omega^2} \left(e^{-i\omega t} + i\omega t - 1\right) \tag{8}$$

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3. Iterate until self-consistency

Questions?

BWs2 theory

PT2 splits the Hamiltonian into a zeroth order part, $\hat{H}^{(0)}$, and a perturbation, $\hat{V}$, as

$$\hat{H} = \hat{H}^{(0)} + \hat{V}. \tag{10}$$

The Moller-Plesset partitioning takes

$$\hat{H}^{(0)} = f_{pq}\hat{a}_p^\dagger \hat{a}_q, \tag{11}$$

where f_{pq} is the Fock matrix. In contrast, the size-consistent Brillouin-Wigner partitioning takes[**?**]

$$\hat{H}^{(0)} = f_{pq}\hat{a}_p^\dagger \hat{a}_q + \frac{\alpha_1}{8}\left(t_{ik}^{ab(1)}\langle jk\|ab\rangle + t_{jk}^{ab(1)}\langle ik\|ab\rangle - \frac{2}{N}t_{kl}^{ab(1)}\langle kl\|ab\rangle\delta_{ij}\right)\hat{a}_i^\dagger \hat{a}_j + \frac{\alpha_2}{8}\left(t_{ij}^{ac(1)}\langle ij\|bc\rangle\right. \tag{12}$$

where $t^{(1)}$ are the doubles amplitudes of the perturbation theory to first order and α_1 and α_2 are empirical parameters.

Cumulant formulations

1. GW cumulant

$$C_{pp}^{R,\mathrm{GW}}(t) = \sum_{i\nu} \zeta_{pi\nu} \left[e^{-i\Delta_{pi\nu}t} - 1 + i\Delta_{pi\nu}t \right] + \sum_{a\nu} \zeta_{pa\nu} \left[e^{-i\Delta_{pa\nu}t} - 1 + i\Delta_{pa\nu}t \right] \quad (13)$$

where $\zeta_{pi\nu} = \left(\frac{M_{pi\nu}}{\Delta_{pi\nu}} \right)^2 \quad , \quad \Delta_{pi\nu} = \epsilon_p - \epsilon_i + \omega_\nu$ and

$\zeta_{pa\nu} = \left(\frac{M_{pa\nu}}{\Delta_{pa\nu}} \right)^2 \quad , \quad \Delta_{pa\nu} = \epsilon_p - \epsilon_a - \omega_\nu.$

2. Second-order cumulant

$$C_{pp}^{(2)}(t) = \frac{1}{2} \sum_{iab} \frac{\langle pi \,||\, ab \rangle^2}{\left(\epsilon_{pi}^{ab} \right)^2} \left(e^{-i\epsilon_{pi}^{ab}t} + i\epsilon_{pi}^{ab}t - 1 \right) + \frac{1}{2} \sum_{ija} \frac{\langle pa \,||\, ij \rangle^2}{\left(\epsilon_{pa}^{ij} \right)^2} \left(e^{-i\epsilon_{pa}^{ij}t} + i\epsilon_{pa}^{ij}t - 1 \right) \quad (14)$$

where

$$\epsilon_{pi}^{ab} = \epsilon_a + \epsilon_b - \epsilon_p - \epsilon_i, \qquad \epsilon_{pa}^{ij} = \epsilon_i + \epsilon_j - \epsilon_p - \epsilon_a.$$

Cumulant formulations (continued)

1. BWs2 cumulant

$$C_{pp}^{\mathrm{BWs2}}(t) = \frac{1}{2}\sum_{iab}\frac{\langle pi\|ab\rangle^2}{\left(\tilde{\varepsilon}_{pi}^{ab}\right)^2}\left(e^{-i\tilde{\varepsilon}_{pi}^{ab}t} + i\tilde{\varepsilon}_{pi}^{ab}t - 1\right) + \frac{1}{2}\sum_{ija}\frac{\langle pa\|ij\rangle^2}{\left(\tilde{\varepsilon}_{pa}^{ij}\right)^2}\left(e^{-i\tilde{\varepsilon}_{pa}^{ij}t} - 1\right), \quad (15)$$

where

$$\tilde{\varepsilon}_{pi}^{ab} = \tilde{\epsilon}_p - \tilde{\epsilon}_i + \tilde{\epsilon}_a - \tilde{\epsilon}_b, \qquad \tilde{\varepsilon}_{pa}^{ij} = \tilde{\epsilon}_p + \tilde{\epsilon}_a - \tilde{\epsilon}_i - \tilde{\epsilon}_j.$$

Dyson equation rearrangement

Start from the definition of the retarded self-energy

$$\Sigma^R_{pp}(\omega) = \left[G^R_{0,pp}(\omega)\right]^{-1} - \left[G^R_{pp}(\omega)\right]^{-1}. \tag{16}$$

We can rearrange to get

$$\left[G^R_{pp}(\omega)\right]^{-1} = \left[G^R_{0,pp}(\omega)\right]^{-1} - \Sigma^R_{pp}(\omega) \quad \Longleftrightarrow \quad G^R = G^R_0 + G^R_0 \Sigma^R G^R. \tag{17}$$

Expanding to first order in the interaction V gives

$$G^R = G^R_0 + G^R_0 \Sigma^R G^R_0 + \mathcal{O}(V^2). \tag{18}$$